Lecture for 11th week

Assoc. Prof. **Bin LI**

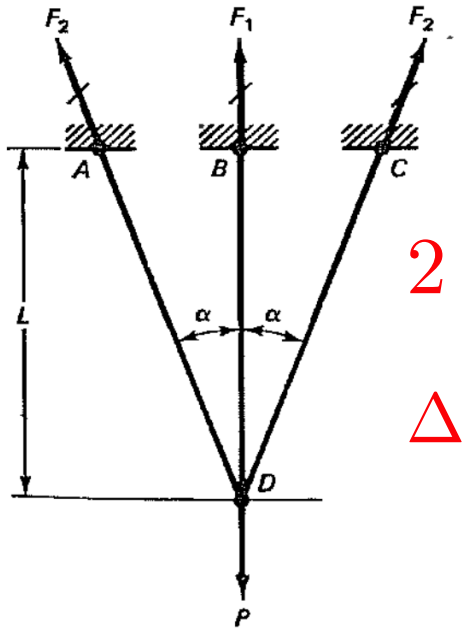
Mechanical Engineering & Robotics,

Guangdong Technion - Israel Institute of Technology

25 May 2026

2. Displacement Method - example

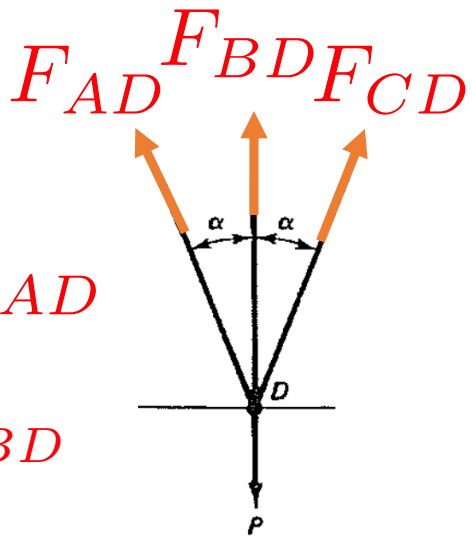
For the planar system of the three elastic bars shown in Fig. 2-47(a), determine the forces in the bars caused by applied force P. The cross-sectional area A of each bar is the same, and their elastic modulus is E.



$$AB^2 + BD^2 = AD^2$$

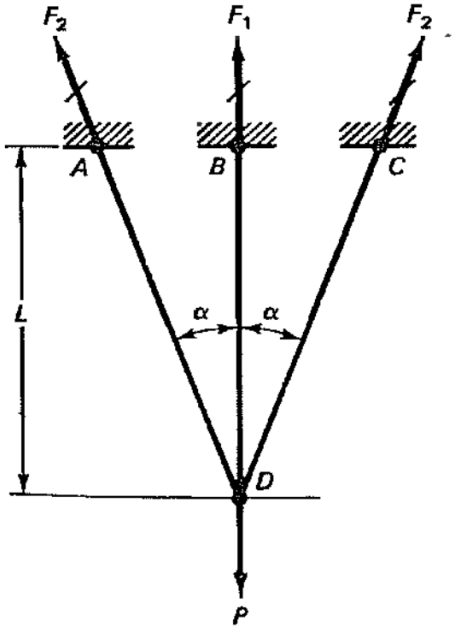
$$2 * BD * \Delta_{BD} = 2 * AD * \Delta_{AD}$$

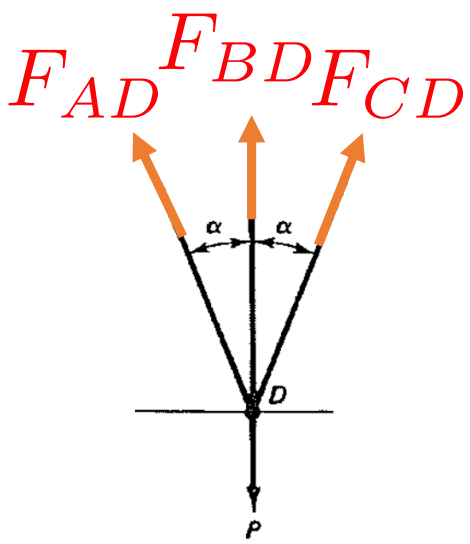
$$\Delta_{AD} = \frac{BD}{AD} * \Delta_{BD} = \cos\alpha * \Delta_{BD}$$



2. Displacement Method - example

For the planar system of the three elastic bars shown in Fig. 2-47(a), determine the forces in the bars caused by applied force P. The cross-sectional area A of each bar is the same, and their elastic modulus is E.





due to symmetry

$$F_{CD} = F_{AD}$$

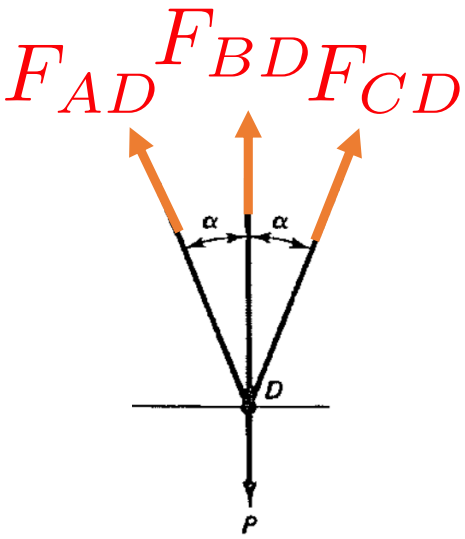
$$F_{CD} = F_{AD} = \Delta_{AD} * \frac{EA}{AD}$$

$$F_{BD} = \Delta_{BD} * \frac{EA}{BD}$$

$$\sum F_y = 0 \Rightarrow \Delta_{BD} * \frac{EA}{BD} + 2\Delta_{AD} * \frac{EA}{AD} * \cos\alpha = P$$

2. Displacement Method - example

For the planar system of the three elastic bars shown in Fig. 2-47(a), determine the forces in the bars caused by applied force P. The cross-sectional area A of each bar is the same, and their elastic modulus is E.



$$\sum F_y = 0 \Rightarrow \Delta_{BD} * \frac{EA}{BD} + 2\Delta_{AD} * \frac{EA}{AD} * \cos\alpha = P$$

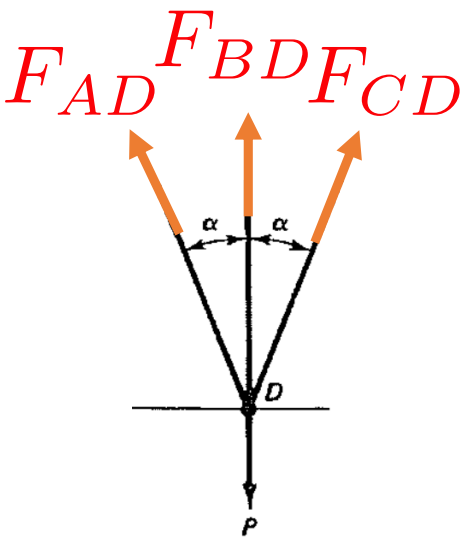
$$\Delta_{AD} = \frac{BD}{AD} * \Delta_{BD} = \cos\alpha * \Delta_{BD}$$

$$\Delta_{BD} = \frac{P}{(1 + 2\cos^3\alpha)\frac{EA}{BD}}$$

$$F_1 = F_{BD} = \Delta_{BD} * \frac{EA}{BD} = \frac{P}{(1 + 2\cos^3\alpha)}$$

2. Displacement Method - example

For the planar system of the three elastic bars shown in Fig. 2-47(a), determine the forces in the bars caused by applied force P. The cross-sectional area A of each bar is the same, and their elastic modulus is E.



$$\Delta_{BD} = \frac{P}{(1+2\cos^3\alpha)\frac{EA}{BD}}$$

$$F_1 = F_{BD} = \Delta_{BD} * \frac{EA}{BD} = \frac{P}{(1+2\cos^3\alpha)}$$

$$\Delta_{AD} = \frac{BD}{AD} * \Delta_{BD} = \cos\alpha * \Delta_{BD}$$

$$F_2 = F_{CD} = F_{AD} = \Delta_{AD} * \frac{EA}{AD} = \frac{P\cos^2\alpha}{(1+2\cos^3\alpha)}$$

2. Exercise

The truss ABC is composed of two members with the same elastic modulus and cross-sectional area, E and A, respectively. The length of member AB is L. The weight of the members is negligible. The weight P is hanged from joint A. What is the minimal weight P_0 that makes point A contact the wall?

